



RAJALAKSHMI
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ME23312 ENGINEERING THERMODYNAMICS

Unit I – Basics, Zeroth and First Law · Presentation 9 of 10

The First Law for Open Systems – the Steady Flow Energy Equation

Control volume and control surface · flow work · assumptions of steady flow · derivation of the SFEE · meaning and magnitude of the terms · closed system versus open system

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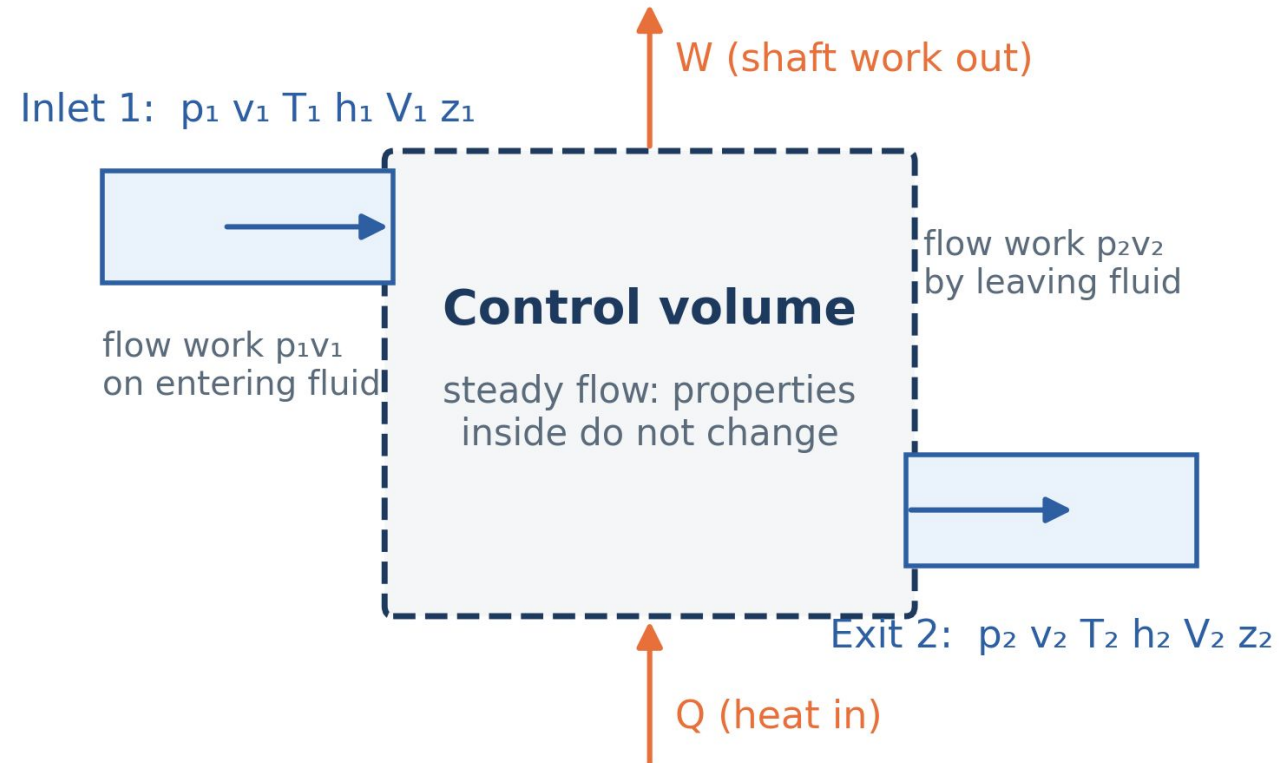
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Control volume and flow work

Open systems and the control volume

- Turbines, compressors, pumps, boilers and nozzles all involve a **continuous flow** of fluid
- Analysed as a **control volume (CV)** – a fixed region in space bounded by a **control surface**
- Mass enters and leaves carrying its energy with it
- Besides heat and shaft work, a new form of work appears: the work required to **push the fluid across the boundary**

The control volume



Fluid enters at 1 and leaves at 2; heat Q and shaft work W cross the control surface

Flow work

- To push mass dm (volume $dm \cdot v_1$) into the CV against pressure p_1 , the fluid behind it does work $(p_1 A_1)(dx_1) = p_1 v_1 dm$
- **Flow work = $p v$ per kg** of fluid at the section
- Fluid leaving at section 2 does flow work $p_2 v_2$ on the fluid ahead of it

Flow work is not a property of the fluid

Flow work, pv per kg, is the work associated with pushing fluid across a boundary – it appears only in open systems.

- It is why the enthalpy $h = u + pv$, and not u , is the natural energy variable for flow processes

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Assumptions of steady flow

Assumptions of steady flow

1. Mass flow rate constant, same at inlet and exit: $\dot{m}_1 = \dot{m}_2 = \dot{m}$
2. State and velocity at inlet and exit do not change with time;
uniform across each section
3. Rates of heat transfer $\dot{Q}$ and work transfer $\dot{W}$ are constant
4. No property inside the control volume changes with time
5. Single inlet and single exit

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Derivation of the SFEE

Energy balance for the control volume

- Apply the first law over a small time dt in which mass dm enters at 1 and an equal mass dm leaves at 2
- Each kg of fluid carries internal energy u , kinetic energy $V^2/2$ and potential energy gz
- **Energy in with fluid + flow work on entering fluid + heat added = energy out with fluid + flow work by leaving fluid + shaft work delivered**

Per kg of fluid flowing

$$u_1 + p_1 v_1 + V_1^2/2 + gz_1 + q = u_2 + p_2 v_2 + V_2^2/2 + gz_2 + w$$

- The combination $u + pv$ appears at each section – this is the **enthalpy h**
- The flow work is absorbed into the enthalpy

The steady flow energy equation (SFEE)

$$h_1 + V_1^2/2 + gz_1 + q = h_2 + V_2^2/2 + gz_2 + w$$

(per kg; h in J/kg, V in m/s, z in m)

- Rate form (multiply by $\dot{m}$): $\dot{Q} + \dot{m} (h_1 + V_1^2/2 + gz_1) = \dot{W} + \dot{m} (h_2 + V_2^2/2 + gz_2)$

Continuity equation

$$\dot{m} = A_1 V_1 / v_1 = A_2 V_2 / v_2 \quad (\text{kg/s})$$

- Accompanies the SFEE and links pipe areas, velocities and specific volumes
- Used to find an exit velocity from the area, or a pipe diameter from the flow rate

Meaning of the terms

Term	Meaning
h	Enthalpy – stored internal energy plus the flow work needed to push the fluid across the boundary
$V^2/2$	Kinetic energy per kg
gz	Potential energy per kg above a datum
q	Heat transfer to the fluid per kg flowing, $\dot{Q}/\dot{m}$
w	Shaft work delivered by the fluid per kg flowing, $\dot{W}/\dot{m}$

Units and magnitude of the terms

- With h in kJ/kg, write the kinetic-energy term as **$V^2/2000$ kJ/kg** and the potential-energy term as **$gz/1000$ kJ/kg**
- A velocity change from 45 to 50 m/s changes KE by only 0.24 kJ/kg; a 10 m rise changes PE by 0.098 kJ/kg
- Enthalpy changes are typically hundreds of kJ/kg
- KE matters only at high velocities (nozzles, jets, exhausts above about 100 m/s); PE only for large elevation differences

Closed system versus steady-flow open system

Closed system

- $Q - W = \Delta U$ (plus ΔKE , ΔPE if the system moves)
- Changes between **initial and final states**
- Energy variable: internal energy u

Open system (SFEE)

- Enthalpy h replaces u – because of the flow work pv
- Changes between **inlet and exit**
- KE and PE of the **flowing stream** appear explicitly

Summary

- Open system = control volume; mass carries energy across the control surface
- Flow work = pv per kg; it turns u into $h = u + pv$
- Steady flow: no accumulation of mass or energy
- SFEE: $h_1 + V_1^2/2 + gz_1 + q = h_2 + V_2^2/2 + gz_2 + w$
- Continuity: $\dot{m} = AV/v$
- $V^2/2000$ and $gz/1000$ in kJ/kg – usually small next to Δh

Next: Applications of the SFEE and unsteady flow

- Nozzle, diffuser, turbine, compressor, boiler, condenser, heat exchanger, throttling valve
- Charging and discharging of vessels
- Numerical problems: Worked Examples 1.6 – 1.11 in the lecture notes